AI – Based Tourism Pressure and Ecosystem Stress Management

- **Kedarnath**
- **Badrinath**
- **Gangotri**
- **Yamunotri**

AVAILABLE DATASETS

A. Tourist Footfall (Season / Month-wise)

Sources

- Uttarakhand Tourism Development Board (UTDB)
- District Tourism Offices
- Char Dham Yatra official reports

Data Type

- Monthly / yearly pilgrim counts
- Peak-season vs off-season trends
- Event-based spikes (Yatra opening, festivals)

B. Satellite & GIS Data

Sources

- **Landsat (USGS)** – long-term forest change (2000–present)
- **Sentinel-2 (ESA)** – high-resolution land-use mapping

Extractable Features

- Forest cover loss
- Urban / road expansion
- NDVI (vegetation stress)

C. Climate Data

Sources

- NASA POWER Climate Dataset
- WorldClim

Variables

- Monthly rainfall
- Max / Min / Mean temperature

Use

- Climate-driven stress vs tourism-driven stress comparison
- Regression & correlation analysis

D. Waste Consumption

Sources

- Swachh Bharat Mission reports
- Research papers on pilgrimage waste

Variables

- Solid waste generation
- Sewage pressure indicators

Use

- Tourism–waste correlation
- Infrastructure stress estimation

PROJECT STAGES

◆ **STAGE 1: Data Collection**

- Collect tourism, climate, waste datasets
- Data cleaning & normalization

- Exploratory Data Analysis (EDA)
- Satellite data sourcing
- GIS boundary mapping (Char Dham regions)
- Initial feature extraction

🔹 STAGE 2: GIS & Feature Engineering

- Forest cover change detection
- Land-use classification (LULC)
- NDVI-based ecosystem stress metrics
- Feature correlation analysis

🔹 STAGE 3: Modeling & Prediction

- ARIMA & LSTM for tourist footfall forecasting
- Climate–tourism interaction modeling
- Regression & feature importance analysis
- Ecosystem stress prediction model
- Threshold identification

🔹 STAGE 4: Dashboard & Early Warning System

- Interactive dashboard (Power BI / Streamlit)
- Risk alerts (High / Medium / Low stress)
- Visual maps & trend charts

◆ **STAGE 5: Real-Time Data Integration**

Real-Time Sources

- Weather APIs
- Crowd density (mobile / proxy data)
- Smart waste & water sensors (simulated if unavailable)

Steps

1. API-based ingestion
2. Live model updates
3. Near-real-time stress alerts